



TAFLOTAN[®]-S ophthalmic solution 0.0015%

(single-dose container)



1. Name of the Medicinal Product

- 1.1 Strength
Tafluprost 0.0015%
- 1.2 Pharmaceutical Dosage Form
Ophthalmic Solution

2. Quality and Quantitative Composition

- 2.1 Qualitative Declaration
Chemical name: 1-Methylethyl (5Z)-7-[(1R, 2R, 3R, 5S)-2-[(1E)-3,3-difluoro-4-phenoxy-1-butenyl]-3,5-dihydroxycyclopentyl]-5-heptenoate
Molecular Formula: C₂₅H₃₄F₂O₅
Molecular Weight: 452.53
Tafluprost occurs as a colorless to light yellow, viscous liquid. It is very soluble in ethanol, diethyl ether and acetonitrile, and practically insoluble in water.
- 2.2 Quantitative Declaration
Each mL contains 15µg of tafluprost.
Each container (0.3mL) contains 4.5µg of Tafluprost.

3. Pharmaceutical Form

TAFLOTAN-S ophthalmic solution is a clear, colorless, sterile aqueous ophthalmic solution.

4. Clinical Particulars

- 4.1 Therapeutic Indication
Reduction of intraocular pressure in patients with primary open-angle glaucoma or ocular hypertension.
- 4.2 Posology and Method of Administration
Instill 1 drop in the affected eye(s) once daily.
- 4.3 Contraindication
1) Patients with combination with products containing omidenepag isopropyl.
2) Patients with a history of hypersensitivity to any of the ingredients of this product.
- 4.4 Special Warning and Precautions for use
- 4.4.1 Careful administration (This product should be administered with care to the following patients.)
1) Patients with aphakia or pseudophakia [This product may induce macular oedema including cystoid macular oedema, and the associated visual acuity reduced.]
2) Patients with bronchial asthma or a history of bronchial asthma [this product may aggravate or induce asthmatic attack.]
3) Patients with endophthalmitis (iritis, uveitis) [other drugs in this category have been reported to cause elevation of intraocular pressure.]
4) Pregnant, parturient and lactating women [see “4.6 Pregnancy and Lactation”.]
- 4.4.2 Important precautions
1) Pigmentation in iris and eyelid (increased melanin content), or hypertrichosis around the eyes may occur. These symptoms gradually progress with continued administration, and stop when the treatment is discontinued. The symptoms like blepharal pigmentation and hypertrichosis around the eyes can gradually disappear or diminish after the administration is discontinued, however, there are reports of the cases that the symptom of iris pigmentation persists even after the administration was discontinued. In such cases, iris color change can be detected clearly in patients with mixed-colored irises and even in patients with single-color dark brown irises (seen among most Japanese) as well. The difference in iris color between right and left eyes could be noted particularly in the case of unilateral administration. As long-term observation data about these symptoms are not yet available, doctors are required to closely observe patients through periodic checkups. Patients should be well informed of the possibility of these symptoms and instructed to wipe off any excess solution from the skin around the eye or to wash the face in order to prevent blepharal pigmentation or hypertrichosis around the eyes.
2) Corneal epithelium disorder (superficial punctate keratitis, filamentary keratitis or corneal erosion) may occur during the treatment. Instruct patients to consult a doctor immediately if subjective symptoms including

- smarting pain, itching and eye pain, continue.
- 3) This product should be administered with care because there is no clinical experience in patients with closed angle glaucoma.
- 4.4.3 Use in the elderly
Because physiological function is generally reduced in the elderly, caution should be exercised.

- 4.4.4 Pediatric Use
The safety of this product to low birth weight infants, neonates, infants or children has not been established. (No clinical experience.)

- 4.4.5 Precautions concerning use
1) Do not use more than once daily because more frequent administration may lessen the IOP-lowering effect.
2) Route of administration: For ophthalmic use only
3) At the time of administration: Followings should be instructed to patients.
(1) Be careful not to touch the tip of the container to the eye directly.
(2) Wipe off or wash the face immediately when any excess solution touches the skin around the eye.
(3) When more than one ophthalmic drug is used, at least 5 minutes of intervals should be taken.
(4) Discard 1 or 2 drops before use (in order to eliminate fragments of container possibly contaminated at opening).
(5) Discard any remaining solution immediately after use. TAFLOTAN-S is a sterilized single-use disposable medical product which does not contain any preservative agents for prevention of cross contamination.

4.5 Interaction with Other Medicinal Products and Other Forms of Interactions

Contraindication for concomitant use (not to use as concomitant drug).

Name of drugs	Clinical symptom/Action	Mechanism/ Risk factor
Omidenepag isopropyl	Moderate or severe eye inflammation such as photophobia and iritis are highly frequently seen.	Unknown

4.6 Pregnancy and Lactation

- 1) This product should be used in pregnant women or women who may possibly be pregnant only if the expected therapeutic benefits are judged to outweigh the possible risks associated with the treatment. [The safety of this product for use during pregnancy has not been established. In animal studies, when tafluprost solution was administered intravenously to pregnant rats at a dose of 30µg/kg/day (2000 times the clinical dose*), teratogenicity and post-implantation embryonic mortality rate increased; at 10µg/kg/day (about 670 times the clinical dose*) adverse effect on fetal development (low body weight and unossification of breast bone in fetuses) was observed. In an intravenous administration in pregnant rabbits at 0.1µg/kg/day (about 6.7 times the clinical dose*), miscarriage and mortality rate after implantation increased, and luteal body and implantation decreased; at 0.03µg/kg/day (2 times the clinical dose*) teratogenicity was observed. In an intravenous administration study in pregnant and lactating rats at a dose level of 1µg/kg/day (about 67 times the clinical dose*), mal-nursing of dams was observed and 4-day survival rate of new born baby decreased. On the other hand, in the study using uteri isolated from rats, uterine contraction was observed at about 3.3 times the plasma concentration of tafluprost (less than 30 pg/mL), or about 420 times the plasma concentration of unbound tafluprost (less than 0.24 pg/mL), calculated based on protein binding ratio, estimated after ocular administration of the clinical dose.]
*Dosage (0.015µg/kg/day) when one drop (30µL) of this product is instilled into both eyes at a time for a 60 kg patient.
- 2) Avoid administration to nursing mothers. If administration is judged to be essential, the patients should be instructed to stop breast-feeding during the treatment. [A study in rats has shown excretion of tafluprost in breast milk after ocular instillation.]

4.7 Effects on Ability to Drive and Use Machine

Temporary blurred vision may develop after administration of this product. Patients should be instructed to refrain from activities like driving or operating machines until the symptom disappears.

4.8 Undesirable Effects

Following safety data were based on the clinical study result of tafluprost ophthalmic solution 0.0015% containing benzalkonium chloride.

Upon Approval

Adverse drug reactions (including abnormal change in laboratory test values) were reported in 326 of 483 patients (67.5%) in clinical studies in Japan. The major adverse drug reactions were conjunctival injection in 151 patients (31.3%), abnormality in eyelashes in 93 patients (19.3%), itching in 85 patients (17.6%), eye irritation in 65 patients (13.5%), iris pigmentation in 39 patients (8.1%) , etc.

Post-marketing surveillance (Upon issuance of the 5th Periodic Safety Report)

Adverse drug reactions were reported in 396 of 3,260 patients (12.1%) in post-marketing surveillance. The major adverse drug reactions were blepharal pigmentation in 93 patients (2.9%), conjunctival injection in 74 patients (2.3%), corneal epithelium disorder including corneal erosion in 58 patients (1.8%), hypertrichosis of eyelids in 40 patients (1.2%), abnormality in eyelashes in 39 patients (1.2%), etc.

1) Clinically significant adverse drug reactions

Iris pigmentation (8.1%): Since iris pigmentation may occur, patients should be examined periodically, and administration should be discontinued depending on the clinical status when iris pigmentation is observed.

2) Other adverse drug reactions

If an adverse drug reaction is observed, appropriate measures including discontinuing administration should be taken.

Incidence Type	Incidence unknown	≥5%	5%> ≥1%	1%> ≥0.1%
Ophthalmic	Conjunctivitis, iritis, keratoconjunctivitis sicca, deepening of upper eyelid sulcus, macular oedema	Conjunctival injection, abnormality in eyelashes (increased length, thickness and number of lashes, etc.), itching, irritation, foreign body sensation, blepharal pigmentation, corneal epithelium disorder including superficial punctate keratitis, abnormal sensation in the eye (discomfort, sticky sensation, dry sensation, etc.)	Eye pain, hypertrichosis of eyelid, eye discharge, photophobia, heavy feeling of eye, lacrimation, blurred vision, conjunctival oedema, blepharitis (redness of eyelid, oedema, etc.)	Subconjunctival haemorrhage
Neuropsychiatric	—	—	Headache	Dizziness
Hypersensitivity	Eyelid dermatitis, rashes	—	Erythema	—
Others	—	—	Elevated AST(GOT), positive urine protein, elevated serum potassium	Elevated ALT(GPT), elevated γ-GTP, positive urine sugar, increased eosinophils, decreased leukocyte count, increased uric acid

Incidence was calculated based on the clinical study results up to the approval of tafluprost ophthalmic solution 0.0015% containing benzalkonium chloride.

4.9 Overdose and treatment

Overdose is unlikely to occur after ocular administration. If overdose occurs, treatment should be symptomatic.

5. Pharmacological Properties

5.1 Pharmacodynamic Properties

1) Intraocular pressure (IOP) lowering effect

Ocular administration of tafluprost ophthalmic solutions ranged from 0.00002% to 0.005% in a single dose to monkeys showed the IOP lowering effect in a concentration-dependent manner. The effect was statistically significant in the groups of the concentration of 0.0005% and higher, compared with the vehicle group. In a repeated dose study in monkeys of tafluprost ophthalmic solutions ranged from 0.001% to 0.005% once daily for 5 days, IOP lowering effect at every concentration was stable and did not attenuate during the administration period.

2) Mechanism of action

Tafluprost acid form, an active metabolite, showed high affinity for the prostanoid FP receptor (Ki=0.40 nM). Aqueous humor dynamics in monkeys was evaluated using fluorophotometry, two-level constant pressure perfusion and ¹²⁵I-¹³¹I labeled albumin perfusion methods following the repeated administration of 0.005% tafluprost ophthalmic solution once daily for 3 to 5 days. Uveoscleral outflow significantly increased without any change in aqueous production.

3) Effect on ocular blood flow

- (1) A repeated instillation of 0.0015% tafluprost ophthalmic solution into rabbit eyes once daily for 28 days significantly increased the blood flow in the optic nerve head, measured with laser speckle method.
- (2) A single dose instillation of 0.0015% tafluprost ophthalmic solution into eyes of healthy adults significantly increased the blood flow rate in the paraoptic nerve head retinal artery and the blood flow at the paraoptic nerve head retina.

5.2 Pharmacokinetic Properties

1) Plasma concentrations

When 0.0025% or 0.005% tafluprost ophthalmic solution was instilled one drop once daily for 7 days in both eyes of 7 healthy adult volunteers, the plasma concentrations of tafluprost and its active metabolite, tafluprost acid form were below the quantitation limits (tafluprost; 0.2 ng/mL, tafluprost acid form; 0.1 ng/mL) at every time point, except in one volunteer of the 0.0025% group at 15 minutes after administration on Day 1, with the concentration of tafluprost acid form being 0.144 ng/mL. (Note: The concentration of this product is 0.0015%.)

2) Ocular tissue distribution in animals

(monkeys)
Following a single ocular instillation of 0.005% ³H-tafluprost ophthalmic solution to monkeys, the radioactivity was distributed in the ocular tissues rapidly and reached the maximum radioactive concentration in the cornea and conjunctiva at 5 to 15 minutes after administration; in the aqueous humor, iris, ciliary body and lens at 2 hours after administration, and was eliminated rapidly.

6. Pharmaceutical Particulars

6.1 List of excipients

Polysorbate 80, concentrated glycerin, disodium edetate hydrate, sodium dihydrogen phosphate dihydrate, sodium hydroxide, hydrochloric acid and purified water

6.2 Incompatibilities

None known.

6.3 Shelf Life

3 years

6.4 Special Precautions for Storage

Store in a refrigerator (2-8°C) and protect from light.
After opening the aluminum foil pouch
Stored below 25°C and protect from light. Use within one month.
Use the product in the pouch which has been opened first.

6.5 Nature and Contents of Container

Transparent polyethylene container
Box of 3 aluminum foil pouch. Each aluminum foil pouch has 10 containers of 0.3 mL. (10 containers per aluminium foil pouch x 3 pouches)

7. Product Registration Holder

SANTEN PHARMA MALAYSIA SDN. BHD.

Unit #23A-10, Q Sentral,
No. 2A, Jalan Stesen Sentral 2, Kuala Lumpur Sentral,
50470 Kuala Lumpur, Malaysia.

8. Manufacturer

SANTEN PHARMACEUTICAL CO., LTD.

Noto plant: 2-14, Shikinami, Hodatsushimizu-cho, Hakui-gun, Ishikawa, Japan.

9. Date of Revision of the Text

January 2022